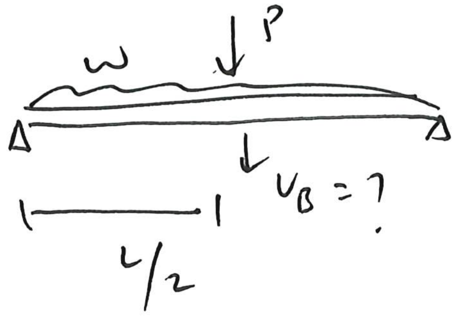
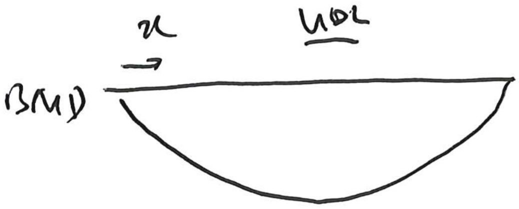
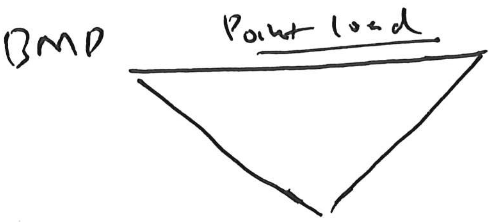
Example 1



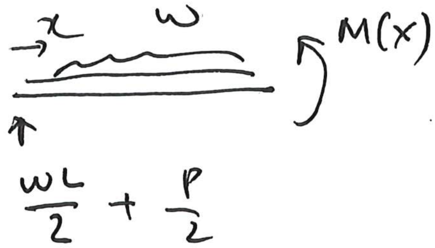
$$V_B = \frac{\partial U}{\partial P_B}, \quad P_B = P$$

$$U = \int_0^L \frac{M^2}{2EI}\,dx$$





$\cancel{M(x)}$



$$M(x) = \frac{wL}{2}x + \frac{P}{2}x - \frac{wx^2}{2}, \quad x \le L/2$$

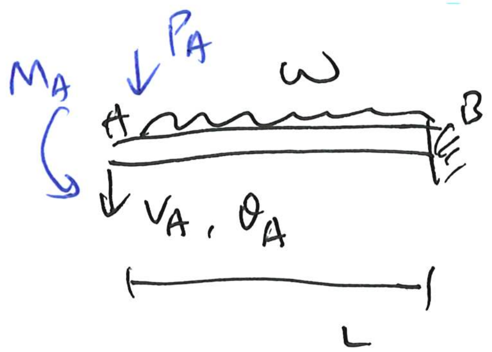
$$\frac{\partial M}{\partial P} = \frac{x}{2}$$

$$\frac{\partial U}{\partial P} = 2\int_0^{L/2} \frac{M}{EI} \cdot \frac{\partial M}{\partial P}\,dx$$

$$= 2\int_0^{L/2} \frac{1}{EI}\left(\frac{wL}{2}x + \frac{P}{2}x - \frac{wx^2}{2}\right)\cdot\left(\frac{x}{2}\right)dx$$

$$= \frac{5wL^4}{384EI} + \frac{PL^3}{48EI}$$

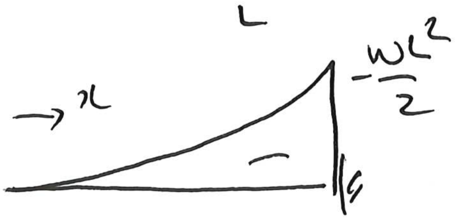
Example 2



Apply dummy loads.

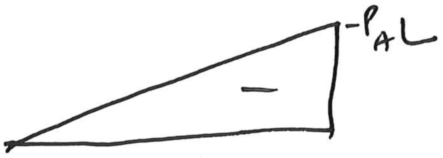
BMD

UDL



$$M_w(x) = -\frac{wx^2}{2}$$

P_A:



$$M_p(x) = -P_A x, \quad \frac{\partial M}{\partial P_A} = -x$$

M_A:

$$M_m(x) = -M_A, \quad \frac{\partial M}{\partial M_A} = -1$$

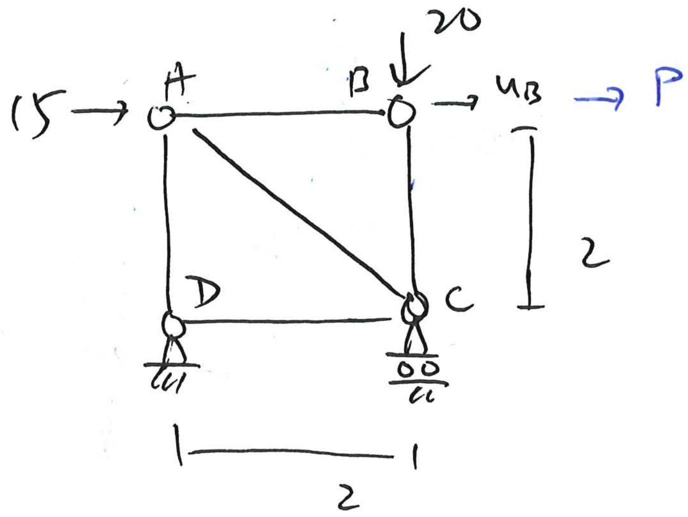
$$V_A = \left.\frac{\partial U}{\partial P_A}\right|_{P_A=0,\, M_A=0} = \left.\int_0^L \frac{M}{EI}\frac{\partial M}{\partial P_A}\,dx\right|_{P_A=0,\, M_A=0}$$

$$= \int_0^L \frac{1}{EI}\left(-\frac{wx^2}{2}\right)(-x)\,dx = \frac{wL^4}{8EI} \quad (\downarrow)$$

$$\theta_A = \left.\int_0^L \frac{M}{EI}\frac{\partial M}{\partial M_A}\,dx\right|_{P_A=0,\, M_A=0}$$

$$= \int_0^L \frac{1}{EI}\left(-\frac{wx^2}{2}\right)(-1)\,dx = \frac{wL^3}{6EI} \quad (\circlearrowleft)$$

Example 3

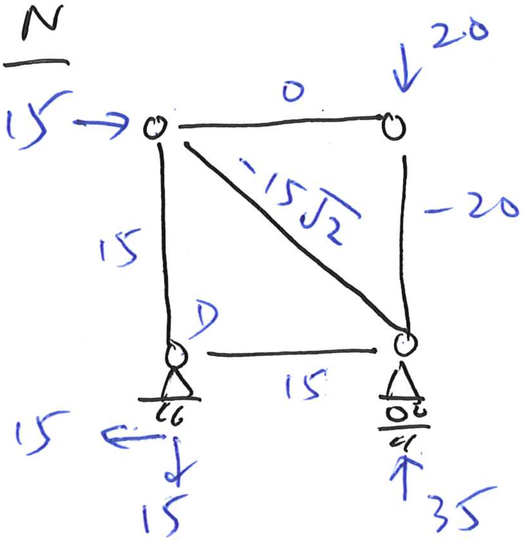


$$u_B = \frac{\partial U}{\partial P}$$

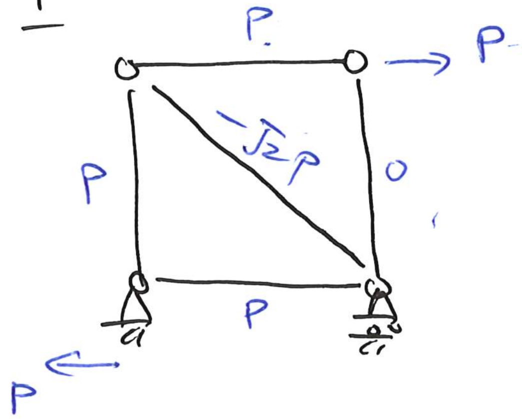
$$U = \sum \frac{N^2 L}{2EA}$$

$$\frac{\partial U}{\partial P} = \sum \frac{NL}{EA} \cdot \frac{\partial N}{\partial P}$$

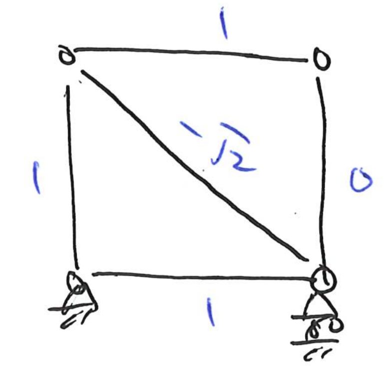
$\underline{N}$



$\underline{P}$



$\frac{\partial N}{\partial P}$



$$u_B = \sum \frac{NL}{EA} \cdot \frac{\partial N}{\partial P}$$

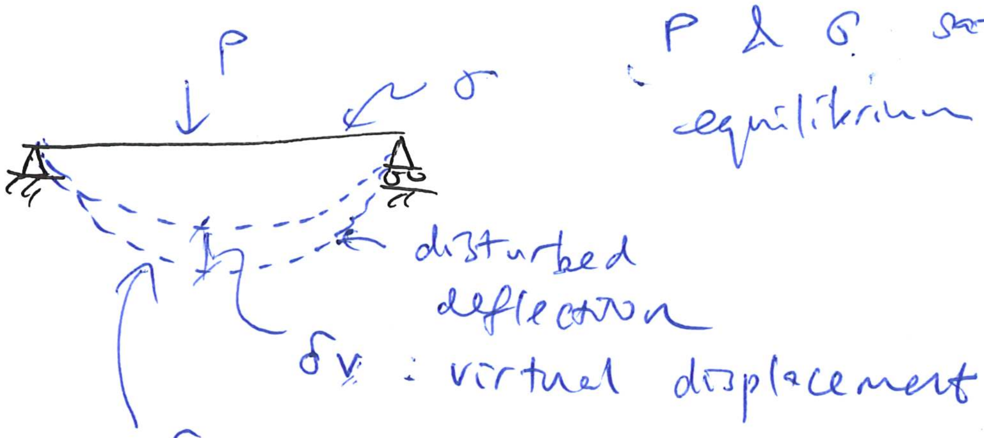
$$= \Big[0 \cdot \frac{2}{EA} \cdot 1 + (-20) \cdot \frac{2}{EA} \cdot 0 + 15 \cdot \frac{2}{EA} \cdot 1$$

$$+ 15 \cdot \frac{2}{EA} \cdot 1 + (-15\sqrt{2}) \cdot \frac{2\sqrt{2}}{EA} \cdot (-\sqrt{2}) \Big]$$

$$\simeq \frac{145}{EA}$$

Principle of virtual work

P & σ satisfy the equilibrium condition



$\delta\varepsilon$: virtual strain

δv & $\delta\varepsilon$ must satisfy the kinematic relation

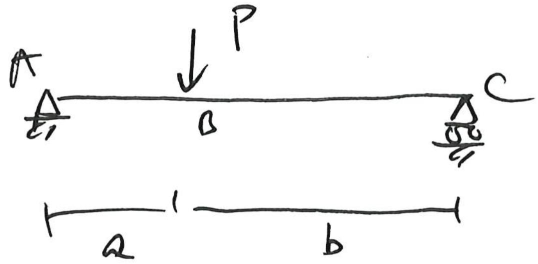
δW : external virtual work

$$= \sum_i P_i \, \delta v_i$$

δU : internal virtual work

$$= \int_V \sigma \cdot \delta\varepsilon \, dV$$

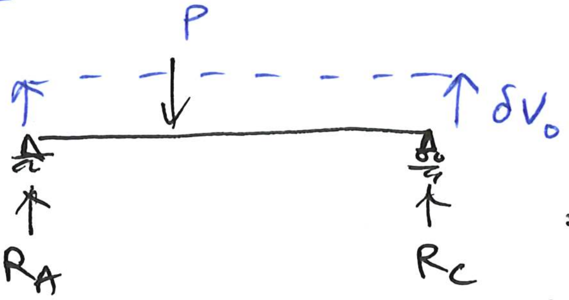
$$\boxed{\delta W = \delta U}$$



$$\delta W = R_A \cdot \delta v_0 + R_C \cdot \delta v_0 + (-P) \cdot \delta v_0$$

$$\delta U = 0 \quad \because \delta \varepsilon = 0$$

δv



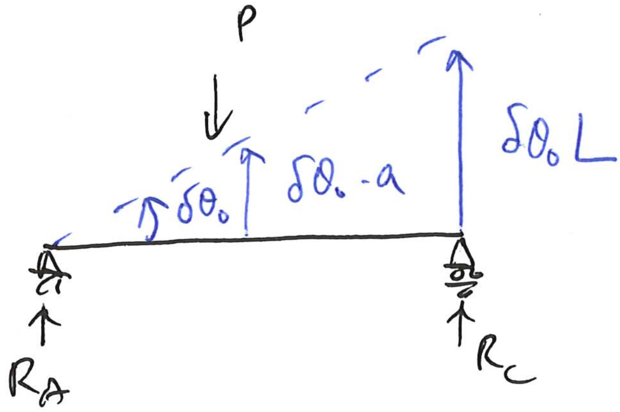
$$\delta W = \delta U = 0$$

$$\Rightarrow (R_A + R_C - P) \cdot \delta v_0 = 0$$

Since δv_0 is arbitrary

$$\Rightarrow R_A + R_C - P = 0 \Leftrightarrow \Sigma F_y = 0$$

δv



~~δW~~

$$\delta W = R_C \cdot \delta\theta_0 L + (-P) \cdot \delta\theta_0 a$$

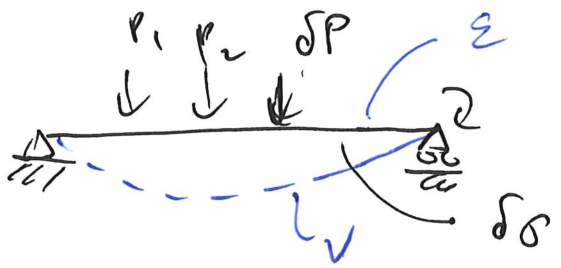
$$\delta U = 0 \quad \because \delta \varepsilon = 0$$

$$\delta W = \delta U \Rightarrow (R_C \cdot L - P \cdot a)\,\delta\theta_0 = 0$$

$\because \delta\theta_0$ is arbitrary

$$\Rightarrow R_C \cdot L - Pa = 0$$

Principle of Complementary Virtual work



v & ε satisfy the kinematic relation

δP & $\delta\sigma$ satisfy the equilibrium condition.

δW : external complementary virtual work

$$= \sum \delta P_i \cdot v_i$$

δU : internal complementary virtual work

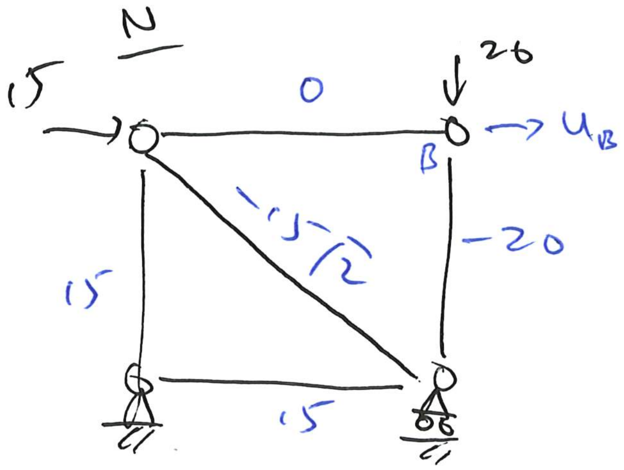
$$= \int_V \delta\sigma \cdot \varepsilon \, dV$$

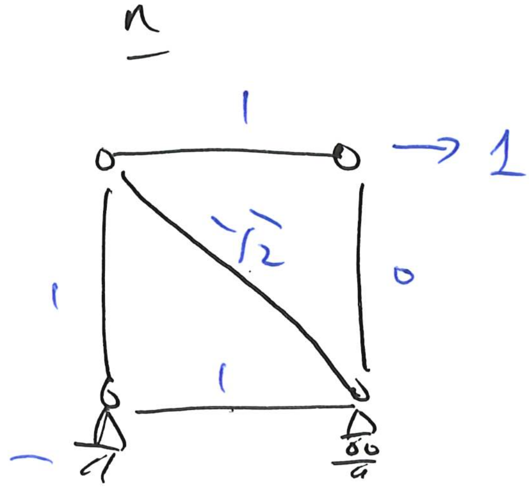
$$\boxed{\delta W = \delta U}$$

aka $\delta W^* = \delta U^*$

linear elastic structures only.

Example 1





$$u_B = \sum \frac{NL}{EA} \cdot n$$

$$= \left(\frac{0 \cdot 1 \cdot 2}{EA} + 0 + \frac{15 \cdot 2}{EA} \cdot 1 + \frac{15 \cdot 2}{EA} \cdot 1 \right.$$

$$\left. + (-15\sqrt{2}) \cdot \frac{2\sqrt{2}}{EA} \cdot (-\sqrt{2}) \right) \simeq \frac{145}{EA}$$